

Cecilia Rosales

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Education

University of California Berkeley (Berkeley, CA)

May 2025

Major: *Electrical Engineering and Computer Sciences, B.S.*

Relevant Coursework: Data Structures and Algorithms, Computer Architecture, Database Systems, Efficient Algorithms, AI, Computer Security, Discrete Math and Probability

Skills

Programming Languages: Python, Java, JavaScript, C

Web: HTML, CSS, Git

Databases: SQLite

Languages: English, Spanish

Projects

Full-Stack 2048 Web Application

- Built a full-stack web application implementing the 2048 game with interactive frontend and backend API integration
- Designed backend endpoint to manage game state and return structured JSON responses consumed by the frontend
- Implemented data flow between frontend UI and backend services using HTTP requests and API calls
- Integrated SQLite database to store and retrieve user scores and leaderboard data
- Built responsive frontend using JavaScript, HTML, and CSS to dynamically render game state
- Utilized: JavaScript, HTML/CSS, SQLite, Git

NGordnet (Data Structures /Backend System)

- Designed backend components for a browser-based data analysis tool for word usage trends
- Built graph-based data structures to process and query large datasets efficiently
- Implemented algorithms to transform raw datasets into structured queryable formats
- Utilized: Java, Git

CS61CPU

- Designed and implemented a virtual CPU that supports assembly-level instructions
- Implemented all the possible assembly language instructions, their variables, and their conditions
- Developed the CPU's functionality such as execution pipeline and register file management
- Utilized: Logisim

Experience

Berkeley Mentorship Cohort – Communications Lead

August 2022 – May 2025

- Design, publish, and update content for the program for 250+ students, both mentors and mentees
- Design and manage social media platforms specific to BMC
- Collaborate with planning and execution of ESS programs and facilitate 10+ events per semester

Clubs/Extracurricular Activities

Hispanic Engineers and Scientist (HES) Garden Committee – Garden Lead

January 2024 – May 2025

- Collaborate on maintaining a garden space on campus during open hours for 3 to 4 hours weekly
- Conduct research on sustainable irrigation methods and the technologies needed to support them
- Raise awareness on sustainable irrigation practices by opening our space to everyone who is interested
- Facilitate 4 to 5 events per semester